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Ecosystem Simulator – Documentation

1. Overview

This project implements a round-based ecosystem simulation using object-oriented Python. An island-like Habitat contains plants and animals that grow, eat, reproduce, and die under environmental constraints such as space, seasons, weather, and food availability.

The simulation is deterministic in structure but uses randomness to model natural variation.

2. Habitat and Simulation Flow

The Habitat manages:

- Available space for ground plants
- Lists of plants and animals
- Seasons, weather, and round progression

Each round executes:

1. Season and environment updates
 2. Plant growth, reproduction, and death
 3. Animal aging, feeding, reproduction, and death
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3. Plant Rules

General Rules

- Plants have a minimum and maximum size, growth rate, space requirement, and lifespan
- Plants die if eaten below their minimum size or if they exceed their maximum age
- Growth depends on seasonal weather and random variation

Plant Types

- **Grass:** Fast growth, drought resistance, emergency reproduction when scarce
- **Tree:** Slow growth, produces Apples every 2 rounds
- **Apple:** No growth, no space usage, decays after a few rounds
- **Eucalyptus:** Can only be eaten by Koalas

Additional Rule – Space Competition

If more than 90% of habitat space is occupied, the smallest plants die first. This models natural selection caused by limited resources.

4. Animal Rules

General Rules

- Animals gain hunger each round based on metabolism

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- Animals must eat regularly to survive
- Death can occur due to starvation, old age, or predation

Feeding Types

- **Herbivores** eat plants
- **Carnivores** hunt weaker animals (never their own species)
- **Omnivores** choose food based on hunger level

Hunting success depends on strength, individual skill, season, and randomness.

5. Reproduction Rules

Animals reproduce only if:

- They are old enough
- Hunger is below a threshold
- Reproduction cooldown has expired

Cooldowns prevent uncontrolled population growth.

6. Animal Welfare (Tierschutz) Rules

To ensure ethical and stable simulation behavior:

1. Anti-extinction protection

- If ≤ 3 animals of a species remain:
 - They cannot die from starvation
 - They cannot be hunted
 - They do not die of old age

2. No cannibalism

- Animals never hunt their own species

3. Seasonal behavior

- Bears hibernate in winter:
 - Reduced hunger
 - No feeding
 - Energy conservation

These rules preserve biodiversity and prevent unrealistic mass extinction.

7. Seasons and Randomness

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- Seasons change every 10 rounds (Spring, Summer, Autumn, Winter)
- Seasons affect plant growth and hunting success
- Randomness influences:
 - Plant growth
 - Hunting outcomes
 - Individual animal traits

Each simulation run is therefore unique.

8. User Handbook (Console UI)

Starting the Simulation

The user defines:

- Habitat size
 - Initial numbers of plants and animals
- All inputs are validated to prevent crashes.

Controls

- **R** – Run one round
- **M** – Run multiple rounds
- **S** – Show ecosystem status
- **Q** – Quit simulation

Output

The status display shows:

- Current round and season
- Space usage
- Detailed lists of plants and animals
- Population summaries by species

9. Assumptions

- No spatial positioning inside the habitat
- Single habitat only
- Focus on clarity and stability over biological precision

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